

ANTHRACITE

Advanced Water Filtration Media

KT[®]



**SUPRIYA WATER
TECHNOLOGY**

Anthracite is a high-rank coal, representing a coal that has been subjected to the highest grade of metamorphism. Anthracite is shiny black, hard and brittle and has the highest fixed-carbon content (approximately 86-98%). Due to its low volatile matter (2–12%), anthracite's combustion process is slow. Most anthracites have low-moisture content (about 3–6%) and their heating value is 34.890 KJ/Kg. Anthracite combusts with hot, clean flame, containing low content of sulfur and volatiles. Due to these characteristics, anthracite is sometimes used in domestic applications or other specialized industrial uses that require smokeless fuels



To successfully utilise anthracite in rotary kilns it is necessary to take special precautions such as grinding the pulverised coal (fuel) to high coal fineness and special burner tip design and high retention time in the calciner are necessary to obtain complete combustion. The radiant nature of rotary kiln refractories and the clinker bed can assist in ignition in much the same manner as radiant arches have done in boilers.

Anthracite is an excellent filter media for water clarification in drinking or industrial use, when used in combination with filtering sands. It is one of the most used filtering media. It is a good complement for the mixed filters, in company of sand or green manganese sand. Due to the special shape of its grains, it allows the suspended particles to be retained in the depth of the filtering bed. Compared to a sand filter, this filtering medium allows a higher flow, less pressure drop and a better and faster backwash.

PHYSICAL PROPERTIES:

○ Color	Black
○ Bulk Density	50 lbs./cu. ft
○ Hardness	3.0-3.8 (Mohs scale)
○ Uniformity Coefficient	<1.7
○ Acid Solubility	<1%

○ Caustic Solubility ((1% NaOH @ 190°F))	<1%
○ Apparent Specific Gravity	1.6 gm/cc
○ Meets AWWA Standard	B100-01
○ Mesh Size	0.6 x 0.8 mm , 0.85 x 0.95 mm, 1.70 x 2.05 mm

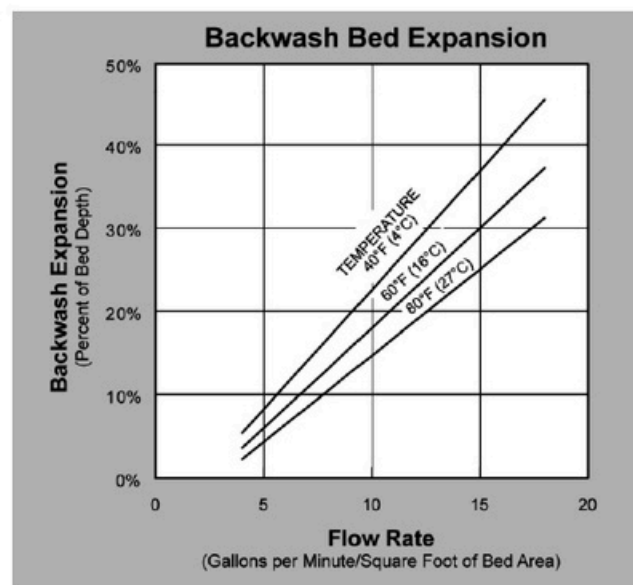
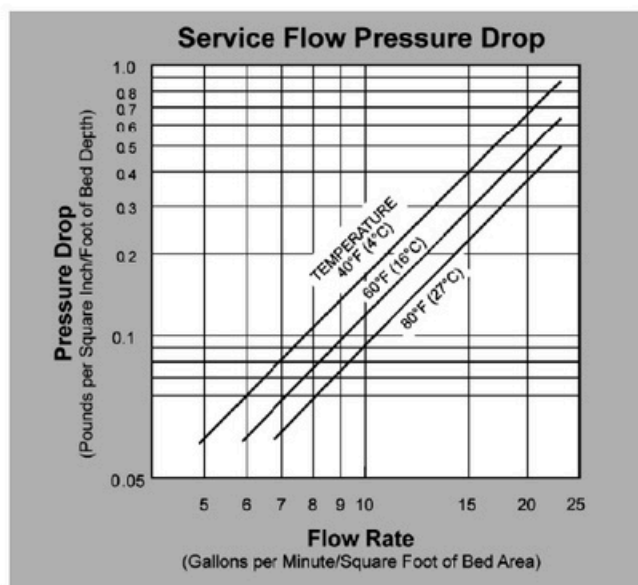
CONDITIONS FOR OPERATION:

▶ Bed depth	24-36 inches
▶ Freeboard	50% of bed depth (min.)
▶ Service flow rate	5 gpm/sq. ft.

▶ Dissolved Oxygen (D.O.) content must be equal to at least 15% of the iron and 29% of the manganese content.	
▶ Backwash flow rate	18-25 gpm/sq. ft

AVERAGE ULTIMATE ANALYSIS (MOISTURE & ASH-FREE BASIS)

Hydrogen	2.1%
Carbon	94.7%
Nitrogen	0.8%
Oxygen	1.6%
Sulfur	0.8%



ADVANTAGES

- △ The shape of particles allows suspended particles to be retained over the depth of the filter bed
- △ It complements mixed filters very well, particularly Green and Manganese Sand filter media
- △ Allows for higher flow rate compared to sand
- △ Smaller pressure drop compared to sand
- △ Faster backwash rate compared to sand
- △ High resistance to a variety of chemicals
- △ Greater physical strength for use in industrial water purification.
- △ Prevents filtration blockage and more.
- △ Prevention of mad ball forming at the surface of the sand layer.
- △ Can easily transform from Single Media Filtration to Dual Media Filtration (sand)

WARNING :

For safety and handling purposes, we recommend appropriate protective measures when entering a wet vessel containing granular anthracite, because wet anthracite depletes oxygen from air and therefore, dangerously low levels of oxygen may be encountered. In such a case, the oxygen level inside the vessel shall be determined before entering and appropriate protective equipment should be worn when entering, or leave the vessel open until the oxygen level in the vessel is normal.

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⚙ We Provide: All Kind of water purification system, Installation of Domestic and industrial RO, 6th Stage of Water Tank Cleaning Service, Media Replacement. Wide range of lighting musical fountains. Official Distributor Of Flavia Alkaline Water Purification System.